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MUKESH PATEL SCH.OF TECH.MGMT ENGG-NM-M

Programme: B. Tech (AI)/B. Tech (AI & DS)/B. Tech (AI & ML)/B. Tech (CE)/B. Tech (CSBS)/B. Tech (CSE - CS)/B. Tech - (CSEDS)/B. Tech (EXTC)/B. Tech (IT)/MBA. Tech (AI)/MBA Tech (Comp)/MBA. Tech (EXTC)/MBA. Tech (IT)

Year: II /Year:III/Semester III/Semester V (Exam Year: 2024-2025)

Subject: Discrete Mathematics

Date: 25 Nov 2024 **Time:** 10:00 am - 01:00 pm (03:00 Hrs.)

Max Marks: 100

FINAL EXAMINATION(2024-2025)

RE EXAMINATION(2023-2024) / (2022-2023)

Instructions:

1. This question paper contains 5 pages
2. Part A is compulsory.
3. Attempt any 4 questions from part B
4. In all 5 questions to be attempted.
5. All questions carry equal marks.
6. Answer to each new question to be started on a fresh page.
7. Figure in right hand side indicates full marks.
8. Assume Suitable data if necessary.

Part A (Compulsory) **20**

i. If $A = \{ x \in \mathbb{N} ; x \leq 10\}$ and we define a relation R on A by $R = \{ (x,y) : x + y \leq 5\}$. Discuss reflexivity, irreflexivity, symmetry and transitivity of the relation R. **4**

ii. Construct the truth table and identify whether statement $(q \wedge p) \vee (q \wedge \sim p)$ is tautology or contingency **4**

iii. If 7 points are chosen inside or on the unit circle, then there is a pair of points that are at a distance at most 1. **4**

iv. a) Draw a graph of $(i)\ K_{3,3}$ $(ii)\ C_5$ **4**

b) Draw the graph for the following adjacency matrix A.

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

v. **4**

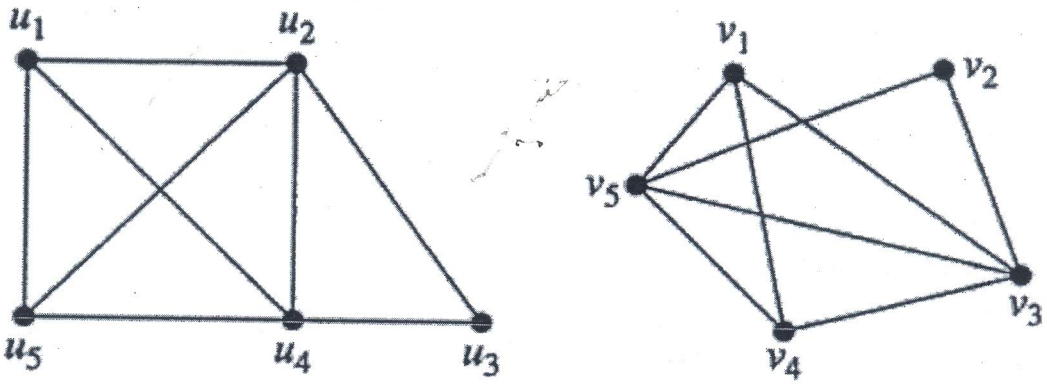
Let $(P(S), \Delta)$ is an abelian group, where S be non-empty set and $P(S)$ be the collection of all subsets of S. Let the binary operation Δ called the symmetric difference of sets be defined as $A \Delta B = (A - B) \cup (B - A)$. Then find identity element of group and inverse of A.

Part B(Attempt Any 4 Questions) **80**

1. **20**

i. Find the general solution of the homogeneous recurrence relation: $a_n = 5a_{n-1} - 6a_{n-2}$ with the initial conditions $a_0 = 2$ and $a_1 = 5$ **6**

ii. Determine whether the following graphs are isomorphic or not. Give reasons. **6**

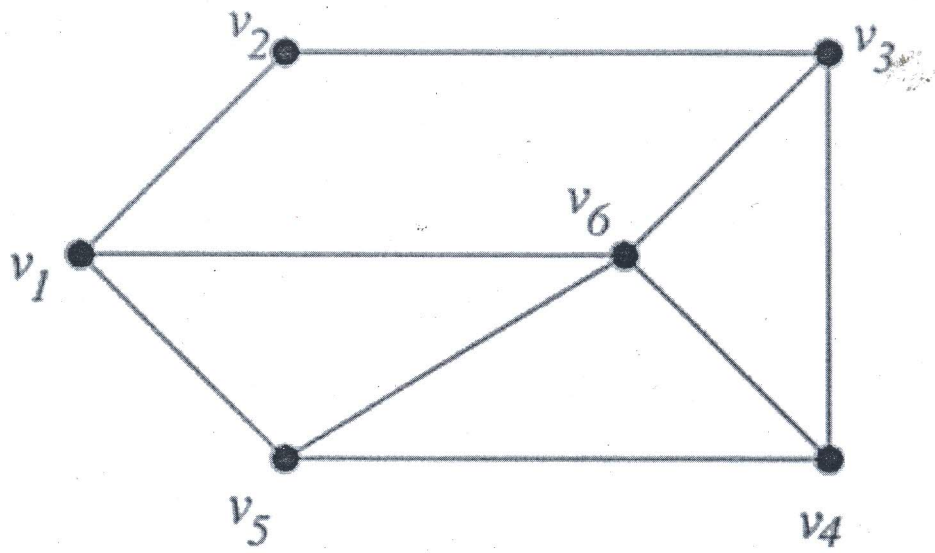


iii. Let R be a relation defined on the set Z of integers given by R = {(a,b) : a+b is an even integer}. Then show that R is an equivalence relation and also give the set of all integers which are related with 3 under the relation R. **8**

2. **20**

i. Prove that $7^n - 1$ is a multiple of 6, using mathematical induction for all positive integers. **6**

ii. Suggest a coloring scheme for the given graph using welsh-powell algorithm. Find the chromatic number for the graph. Justify each step. **6**



iii. In a survey of 260 college students, the data were obtained 64 had taken a mathematics course 94 had taken a computer science course 58 had taken a business course 28 had taken both a mathematics and business course 26 had taken both a mathematics and computer science course 22 had taken both a computer science and business course 14 had taken all three types of courses. **8**

(a) How many students were surveyed who had taken none of three types of courses ?

(b) Of the students surveyed, how many had taken only a computer science course?

3. **20**

i. Show that for a real number α , the set of matrices $A_\alpha = \begin{bmatrix} cos\alpha & -sin\alpha \\ sin\alpha & cos\alpha \end{bmatrix}$ forms an abelian group under usual matrix multiplication. **6**

ii. If you have to access the network, then you can change your grade. You have to access the network. Therefore you can change your grade. **6**

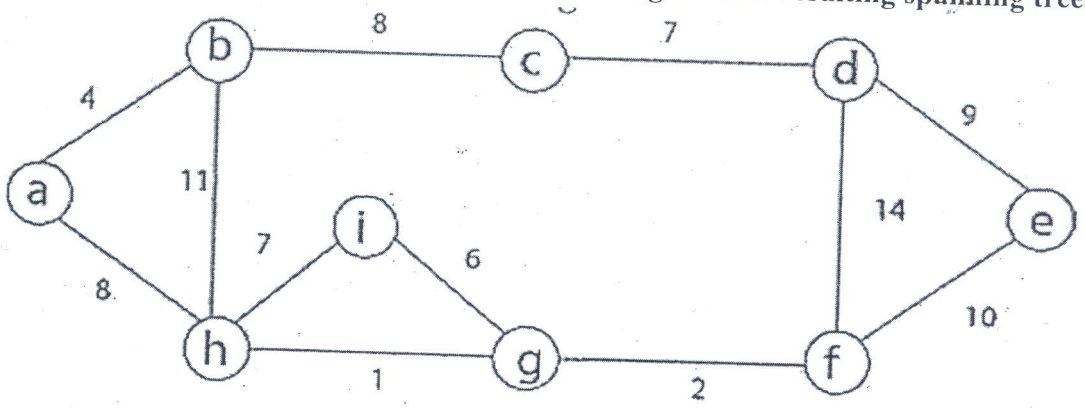
iii. Find the solution of non-homogeneous recurrence relation: $a_n = 5a_{n-1} - 6an - 2 + n$ and $a_0 = 0$ & $a_1 = 2$ **8**

4. **20**

i. Let (G, $*$) be a group and f : G $\rightarrow$ G defined as f(x) = a$*$ x$*$ a^{-1}, x $\in$ G. for fixed a$\in$ G. Then show that f is an isomorphism. **6**

ii. Using technique of contradiction prove that $\sqrt{3}$ is an irrational number. **6**

iii. **Construct a minimum spanning tree for the following graph using Kruskal's Algorithm, and hence find the minimum total weight of the resulting spanning tree.** **8**



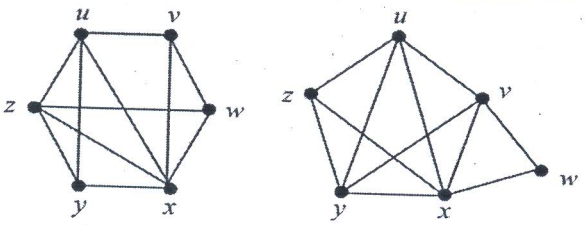
5. **20**

i. $f: R \to R$ is defined by $f(x) = x^3$, $g: R \to R$ is defined by $g(x) = 4x^2 + 1$ $h: R \to R$ is defined by $h(x) = 7x - 2$ **6**

Find the rule defining $(i)\ fog$ $(ii)\ gof$ $(iii)\ (goh)of$ $(iv)\ go(hof)$

Explain the planar graph. Further, determine whether the following graphs are planar. **6**

ii.



G_1 G_2

iii. **Show that $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ is a ring under the usual operations of addition and multiplication .** **8**

6. **20**

1. Obtain disjunctive normal form of: **6**

$$p \longrightarrow ((p \longrightarrow q) \wedge \sim (\sim q \vee \sim p))$$

2. **6**

Show that the set $G = \{1, 2, 3, 4\}$ forms an abelian group with respect to the multiplication modulo 5.

3. Determine if the given graph has a) A Eulerian path b) A Eulerian cycle c) A Hamilton Path d) A Hamilton cycle 8

